

DDoS Incident Analysis — Case Studies

6) Google Cloud — 46 Million RPS Attack (June 2022)

Target:

A Google Cloud Armor customer running a large-scale web service. Google did not disclose the customer's name but stated it was a popular online platform delivering web-based applications to users globally.

Technology Used:

A massive **HTTP/HTTPS Layer-7 flood** peaking at **46 million requests per second (RPS)** — the largest ever recorded at the time. The attack originated from **5,256 source IPs** spanning 132 countries, many leveraging compromised proxies and VPNs to obscure origins. It mimicked legitimate HTTPS requests to evade signature-based filtering.

Attacker Motive:

Likely an **extortion or stress test campaign**. Researchers suspect the actors were experimenting with new botnet orchestration tools to measure the effectiveness of Google's web-layer defense.

Impact:

Due to Google Cloud Armor's adaptive protection, the attack was absorbed without downtime. A less prepared target could experience **total web service unavailability** and **application-layer exhaustion** resulting in prolonged outages.

Defensive Strategies:

- Deploy **application-layer firewalls (WAFs)** with adaptive filtering
- Use **rate-limiting per user/session/token** for high-RPS services
- Implement **progressive challenge-response systems (CAPTCHAs)**
- Analyze **TLS fingerprinting** and user-agent patterns to detect spoofed requests
- Maintain **geo-throttling** and automated source reputation updates

7) AWS — 2.3 Tbps Attack (February 2020)

Target:

An unidentified AWS customer in the **financial technology sector**, relying heavily on AWS infrastructure for API-based services.

Technology Used:

An **amplified UDP reflection attack** exploiting **CLDAP (Connectionless LDAP)** servers. The adversaries coordinated thousands of reflectors to produce short but high-bandwidth bursts that reached **2.3 Tbps**, one of the largest attacks ever recorded by AWS Shield.

Attacker Motive:

Suspected **financial extortion** or **retaliatory attack** due to service disputes. CLDAP-based vectors are typically used by experienced criminal botnet operators testing infrastructure resilience.

Impact:

AWS Shield Advanced automatically mitigated the attack. However, smaller fintech providers connected via peering networks observed transient latency and congestion.

Defensive Strategies:

- Disable or restrict **CLDAP exposure** on public servers
- Use **AWS Shield Advanced** or equivalent tiered DDoS protection
- Implement **ingress filtering (BCP38)** to block spoofed packets
- Maintain **threat intelligence feeds** to block known reflectors
- Simulate high-bandwidth tests to verify mitigation readiness

8) GitHub — 1.35 Tbps Memcached Reflection Attack (February 2018)**Target:**

GitHub.com, the world's largest code hosting platform.

Technology Used:

An **amplification attack** leveraging **misconfigured Memcached servers**. Attackers sent small UDP queries that were amplified up to 50,000× by vulnerable servers, peaking at **1.35 Tbps** of inbound traffic.

Attacker Motive:

Likely **ransom-based disruption** or a **proof-of-concept demonstration** exploiting newly discovered Memcached vulnerabilities.

Impact:

GitHub experienced intermittent outages for about **8 minutes** before Akamai Prolexic's scrubbing service absorbed the load. The incident highlighted how quickly new attack vectors can be weaponized.

Defensive Strategies:

- Disable UDP on Memcached or restrict access to trusted IPs
- Deploy **real-time upstream scrubbing partnerships**
- Maintain **attack vector watchlists** to quickly patch exploited protocols
- Apply **traffic mirroring and flow sampling** to identify emerging anomalies

9) OVHcloud — 1 Tbps IoT Botnet Attack (September 2016)**Target:**

OVHcloud, a major European hosting provider. Attackers targeted OVH's data centers hosting popular game servers.

Technology Used:

A **Mirai-based IoT botnet** using hundreds of thousands of compromised IP cameras and routers. The

attack peaked at **990 Gbps – 1 Tbps** with mixed TCP SYN and ACK floods targeting OVH's gaming infrastructure.

Attacker Motive:

Believed to be **competitive disruption** between gaming server operators. Some evidence suggested involvement of rival hosting resellers.

Impact:

OVH services experienced degraded performance and temporary disconnections. The attack demonstrated the massive scale IoT-based DDoS networks could achieve.

Defensive Strategies:

- Use **network-wide SYN flood protection** and **TCP cookie validation**
- Harden **IoT device firmware** and enforce **ISP-level blocking** of infected IPs
- Implement **multi-layered DDoS filtering** combining volume + protocol defenses
- Encourage **device manufacturers** to adopt security-by-default

10) Dyn DNS — 1.2 Tbps Mirai Botnet Attack (October 2016)

Target:

Dyn, a major DNS provider, which serviced platforms like Twitter, Netflix, Reddit, and Spotify.

Technology Used:

The **Mirai botnet**, composed of ~100,000 IoT devices (cameras, DVRs, routers), launched a massive DNS flood that reached **1.2 Tbps**, disrupting name resolution for major websites across North America and Europe.

Attacker Motive:

Unclear — theories include **hacktivism, revenge, or political protest**. Some reports suggest it was a “stress test” to demonstrate botnet power.

Impact:

Massive service outages lasting several hours. Major online platforms became unreachable. The event was a wake-up call for the global internet community on IoT insecurity.

Defensive Strategies:

- Distribute DNS infrastructure across **anycast networks**
- Implement **DNS query rate limiting** and **response caching**
- Promote **IoT security compliance and manufacturer accountability**
- Maintain **redundant DNS providers** for business continuity